Model Evaluation Plan-Combining Reports Fundamental Data in Stocks Prediction Models (Capstone Project)

This document outlines the comprehensive evaluation framework used in the Stock AI Robo-Advisor project, integrating time-series models and fundamental financial data extracted from company reports.

Data Sources

1) **Stock Price Data-** Daily “OHLCV” data (Open, High, Low, Close, Volume) is taken from Yahoo Finances
2) **Financial Reports-** Quarterly company data (Revenue, Net Profit, ROE, EPS, etc.) are extracted from both TASE’S MAYA interface and Yahoo Finance API.

Stock EDA & Feature Engineering

Before modeling, a structured EDA process is applied for each ticker:

- **Missing Handling**: Failed tickers are logged; NaN imputation via .bfill() and .ffill().
- **Feature Engineering**:
  - Rolling means: 5d, 10d, 20d of adjusted close.
  - Volatility: 10-day standard deviation.
  - Differences: High–Low and Open–Close.
  - RSI and MACD technical indicators.
  - Binary up/down movement flags.
  - Returns (percentage daily change).
- **Data Aggregation**:
  - Per-ticker groupings
  - Concatenation to a multi-index structure.

Financial Reports Feature Extraction Pipeline

1) reports to json files:

- Parses quarterly financial reports sourced from the TASE MAYA system.
- Utilizes string pattern matching & Regex to extract structured financial metrics from unstructured reports.
- Fields extracted include: Total Revenue, Net Profit, EPS, Total Assets, Equity, Operating Profit, and Cash Flow.
- Extracted values are normalized (e.g., converting "אלפי ש"ח" to numerical values in millions).
- Each extracted entry is labeled with company name, report date, and the corresponding quarter (YYYY-QX format, by the json file name)

2) Json semi-structured data to csv structured data:

- Links raw extracted metrics to official company tickers via handly-made mapping from .xlsx. file
- Produces a consolidated per-quarter dataset ready to be merged with time-series features for modeling.

3) **Correlations between following quarters:**
    - For a given company, computes Pearson correlation between each financial parameter and the company's stock returns over future quarters (lag of 1).

Modeling Framework

Each stock is modeled independently using the following configurations. All notebooks follow the structure of: **data preprocessing → model definition → training loop → evaluation**.

LSTM & GRU (Deep Learning)

Data Sources

- Daily OHLCV stock price data from Yahoo Finance (2020–2024).
- Quarterly company report data extracted from TASE’s MAYA interface and parsed via a custom financial report pipeline, listed above.

Feature Engineering

- **Technical Indicators:**
    - High–Low and Open–Close differences
    - Rolling means (5d, 10d, 20d), volatility (10d std), RSI, MACD, up/down flags, daily returns
- **Financial Injection:**
    - Precomputed correlation values per quarter:
        - metrics change: correlation of financial metric between following quarters.
        - price change: company stock’s price change after 1, 30, 90 days.
        - score: strength of (pearson) correlation signal between financial metric and price movement
    - Injected only at official report dates (e.g., 31-Mar, 30-Jun, etc.)

    - No forward-filling is applied to preserve the sparsity of financial events.
    - This allows the model to learn the lagged effect of quarterly disclosures, as neural models capture temporal dependencies inherently.

Preprocessing

- Technical and financial features scaled separately using RobustScaler.
- Financial features normalized only over real-value entries (ignoring padded zeros).

Model Configuration

- Model Type: LSTM or GRU
- Architecture:
  - 4 recurrent layers: LSTM(100) or GRU(100) with Dropout(0.1)
  - BatchNormalization layer
  - Dense stack: Dense(25) → Dropout → Dense(25) → Dense(1)
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

Evaluation

- 90/10 chronological split (time-based)
- Evaluation metrics: MSE, MAE, MAPE, R^2

Forecasting

- 30-day forward simulation using normal-distributed percentage changes based on recent predictions

Output Summary

- One .h5 model per ticker
- CSV files containing:
    - Actual vs Predicted values (test set)
    - 30-day forecasted values
    - Evaluation metrics

XGBoost & LightGBM (Boosted Trees)

Data Sources

- Daily OHLCV stock price data from Yahoo Finance (2020–2024)
- Quarterly financial report data extracted from TASE's MAYA system, aligned to quarterly timestamps

Feature Engineering

- **Technical Indicators:**
    - Rolling mean, standard deviation, and median for windows of 2, 5, 10, 20, and 60 days
- **Financial Injection:**
    - Precomputed per-quarter correlation data:
        - metrics_change, price_change, score for Net Income, Revenue, Total Assets
    - Injected only at official report dates (e.g., 31-Mar, 30-Jun)
    - Forward-filled only if non-zero:
        - Avoids propagating fake signals
        - Enables financial context to persist between events, which is important for models that do not retain memory (like XGBoost)

Preprocessing

- Features normalized using StandardScaler
- Target: Raw closing price of the stock

Model Configuration

- Model Type: XGBRegressor or LGBMRegressor
- **Hyperparameters:**
    - n_estimators = 100
    - learning_rate = 0.05
    - max_depth = 6
    - random_state = 42
- Each stock is modeled and saved independently

Evaluation

- 90/10 chronological split (non-shuffled)
- Evaluation metrics: MSE, MAE, MAPE, R^2

Forecasting

- 30-day forward forecast using normal-distributed percentage changes sampled from test predictions to simulate price movement volatility

Output Summary

- One .pkl model per stock
- CSV files containing:
    - Actual vs Predicted values (test set)
    - 30-day forecasted values
    - Evaluation metrics

Model Evaluation Metrics

- **ΔMAPE**: Measures the improvement (or lack thereof) when including financial report features versus time-series-only models. **The MAPE difference will be our main metric differing the experiment results, answering the most important question- Are report parameters contributing to stocks prediction?**
- **MAPE (Mean Absolute Percentage Error)**: Used as the primary metric for evaluating relative forecasting accuracy, especially useful for financial data due to its scale-invariance.
- **R^2 (Coefficient of Determination)**: Measures how well the predicted values explain the variance of the actual target values. Useful for detecting underfitting.
- **MAE / MSE (Mean Absolute Error / Mean Squared Error)**: Provide robust support metrics to evaluate prediction error magnitude, with MSE penalizing larger errors more heavily.
- **Negative R^2 Count**: Tracks how many stocks had $R^2 < 0$, indicating models that failed to capture even baseline variance.

Experimental Settings

- Two experimental conditions:
    1. **Baseline**: OHLCV time-series only.
    2. **Extended**: OHLCV + report parameters.
- Each model is trained per condition for all available stocks.
- Per-ticker selection of best-performing model (lowest MAPE).

Output Summary

Summarizing our experiments, with and without the financial report parameters, we would like to examine our main experiment question from different aspects, leading to a deeper understanding:

- Per-stock CSVs of actual vs. predicted values & forecasted values for the next 30 days in each of the experiments.
- Integration into Streamlit chatbot with Gemini analysis
- Model performance tables and ΔMAPE plots